

Shop test protocol

PRELIMINARY

Engine type	12V175DML	Building standard A10
Number of cylinders	12	in vee angle
Bore	175	mm
Stroke	215	mm
Engine number	8351446	
Feature ident	A10-MFAA2D109A	
Engine output	2.400	kW
Engine speed	2.000	rpm
Engine rotation direction	Counter clockwise	
Firing order	A1 - B2 - A2 - B4 - A4 - B6 - A6 - B5 - A5 - B3 - A3 - B1	
Application	4 - stroke	
	Marine main engine	
	FPP: Fixed pitch propeller	

All tests are carried out according to ISO 3046-1, -3 and ISO 15550.

Classification society	LR
Emission-Certification	NOx-Emission according IMO-Regulation 13 revised MARPOL Annex VI 2008 IMO-Tier II-Certification engine group (parent)

Emission test cycle E3

Classification Engine	CPN2330412-2
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Customer	Lüssen Ship Yard
Place / Country	Germany

Place / Country	Frederikshavn / Denmark
Date	21-09-2023

Engine type 12V175DML

Engine no. 8351446

Representatives

Name	Company
Søren Sørensen	LR
Kim Sørensen	MAN Energy Solutions SE
Jonas Nørgaard	MAN Energy Solutions SE
Tobias M. Hübener	Lürssen Ship Yard
Calvern Newby	Owner Representation

Certification

This is to certify that the shop test for the above mentioned engine has been carried out according to the enclosed recordings and performance data.

Representative of classification society

Søren Sørensen



Representative of customer

Tobias M. Hübener 21.09.2023

Representative of owner

Calvern Newby

Representative of MAN Energy Solutions SE

Kim Sørensen

Representative of MAN Energy Solutions SE

Jonas Nørgaard

Place, date

Frederikshavn, 21-09-2023

Engine no. 8351446

ne no. 8351446

PRELIMINARY

as.- D.	Remark

[illegible]

FAT 25%

Performance Data



21-09-2023, 10:10

25,0

%

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	23,2	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,42	g/kg
turbocharger no.	1343-02/-03	relative humidity	63	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990	kJ/kg

Power

engine power	599,9	kW
engine speed	1.259	rpm
mean eff.press.	9,2	bar

Fuel

fuel temp. HP pump inlet	37	°C
fuel press. HP pump inlet	7,7	bar
fuel mass flow 1	124,1	kg/h
s.f.o.c (42700kJ/kg,ISO)	206.9	g/kWh

Common Rail

fuel index	45,4	%			
current feed duration A MV	1,1	ms	current feed duration B MV	1,1	ms
current feed timing A MV	5,0	°KW	current feed timing B MV	5,0	°KW

Charge Air

charge air temp. A	39,3	°C	air temp.compr.inlet B	38,9	°C
			charge air temp. B	38,7	°C
			charge air press. B	450	mbar

Exhaust Gas

	MV	cyl.	1	2	3	4	5	6
temp. cylinder outlet (A)	- °C	A	-	-	-	-	-	-
temp. cylinder outlet (B)	- °C	B	-	-	-	-	-	-

exh.gas temp.turb inlet A	539,0	°C	exh.gas temp.turb inlet B	538,0	°C
exh.gas press.turb.outlet A	8	mbar	exh.gas press.turb.outlet B	8	mbar
TC.speed A	29.440	rpm	TC.speed B	29.320	rpm
wastegate position A	0	%	wastegate position B	0	%

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT			SW		
cw.press.engine inlet	1,26	bar	sea water press.pump outlet	2,58	bar
cw.temp.engine outlet	86,3	°C	sea water temp.pump inlet	23,8	°C

Bearing

		MV	No.	cs	1	2	3	4	5	6	7	ccs
main bearing temperature	94,7	°C	°C	86,4	93,5	94,8	96,2	93,7	94,8	94,0	95,5	81,7
splash oil temperature	93,7	°C	°C		94,2	93,5	94,9	92,8	92,6	94,2		



Mesa

press.crankcase	-1 mbar	smoke AVL	-	-
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optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 467,5 °C

measurement-no: 61
FAT 48%

Performance Data



47,8 %

21-09-2023, 10:25

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	23,7	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,61	g/kg
turbocharger no.	1343-02/-03	relative humidity	62	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42,990 kJ/kg	

Power

engine power	1.146,4	kW
engine speed	1.459	rpm
mean eff.press.	15,2	bar

Fuel

fuel temp.HP pump inlet	38	°C
fuel press.HP pump inlet	8,0	bar
fuel mass flow 1	223,4	kg/h
s.f.o.c (42700kJ/kg,ISO)	195,3	g/kWh

Common Rail

fuel index	62,6	%		
current feed duration A MV	1,3	ms	current feed duration B MV	1,3 ms
current feed timing A MV	9,6	°KW	current feed timing B MV	9,6 °KW

Charge Air

charge air temp. A	38,8	°C	air temp.compr.inlet B	34,7 °C
			charge air temp. B	38,0 °C
			charge air press. B	1.220 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	571,0	°C								exh.gas temp.turb inlet B 573,4 °C
exh.gas press.turb.outlet A	30	mbar								exh.gas press.turb.outlet B 30 mbar
TC.speed A	44.380	rpm								TC.speed B 44.310 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	1,40	bar
cw.temp.engine outlet	86,5	°C

LT

cw.press.IC LT inlet	1,24	bar
SW		
sea water press.pump outlet	3,14	bar
sea water temp.pump inlet	23,6	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	99,2	°C	°C	88,5	97,3	100,3	101,4	97,6	99,3	98,9	99,3	83,2
splash oil temperature	97,6	°C	°C		98,1	97,0	99,3	96,8	96,3	98,1		



14.6.20

AMS

press.crankcase -2 mbar smoke AVL - -

optional values shown as "-" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 470,1 °C



21-09-2023, 10:45

50,1 %

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	24,2	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,65	g/kg
turbocharger no.	1343-02/-03	relative humidity	60	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	1.201,7	kW
engine speed	1.586	rpm
mean eff.press.	14,6	bar

Fuel

fuel temp.HP pump inlet	39	°C
fuel press.HP pump inlet	8,2	bar
fuel mass flow 1	236,7	kg/h
s.f.o.c (42700kJ/kg,ISO)	197,4	g/kWh

Common Rail

fuel index	61,4	%		
current feed duration A MV	1,3	ms	current feed duration B MV	1,3 ms
current feed timing A MV	10,8	°KW	current feed timing B MV	10,8 °KW

Charge Air

charge air temp. A	38,4	°C	air temp.compr.inlet B	35,3 °C
			charge air temp. B	37,7 °C
			charge air press. B	1.360 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	552,0	°C								exh.gas temp.turb inlet B 550,0 °C
exh.gas press.turb.outlet A	39	mbar								exh.gas press.turb.outlet B 39 mbar
TC.speed A	46.530	rpm								TC.speed B 46.520 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,5	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT			LT		
cw.press.engine inlet	1,52	bar	cw.press.IC LT inlet	1,36	bar
cw.temp.engine outlet	86,2	°C	SW		
			sea water press.pump outlet	3,53	bar
			sea water temp.pump inlet	23,9	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	100,9	°C	°C	89,4	99,1	102,1	103,5	99,0	101,1	100,5	101,2	83,8
splash oil temperature	97,7	°C	°C		97,0	97,5	99,7	97,1	96,8	98,4		



MAN

press.crankcase -3 mbar smoke AVL - -

optional values shown as "" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 448,2 °C

measurement-no: 63
FAT 75%

Performance Data



21-09-2023, 11:00

75,1 %

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	24,5	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,63	g/kg
turbocharger no.	1343-02/-03	relative humidity	59	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	1.801,8	kW
engine speed	1.816	rpm
mean eff.press.	19,2	bar

Fuel

fuel temp.HP pump inlet	37	°C
fuel press.HP pump inlet	8,1	bar
fuel mass flow 1	366,9	kg/h
s.f.o.c (42700kJ/kg,ISO)	204,4	g/kWh

Common Rail

fuel index	80,8	%		
current feed duration A MV	1,7	ms	current feed duration B MV	1,7 ms
current feed timing A MV	10,8	°KW	current feed timing B MV	10,8 °KW

Charge Air

charge air temp. A	37,5	°C	air temp.compr.inlet B	32,2 °C
			charge air temp. B	37,0 °C
			charge air press. B	2.365 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	597,0	°C								exh.gas temp.turb inlet B 600,0 °C
exh.gas press.turb.outlet A	96	mbar								exh.gas press.turb.outlet B 96 mbar
TC.speed A	58.050	rpm								TC.speed B 58.060 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,5	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT			LT		
cw.press.engine inlet	1,73	bar	cw.press.IC LT inlet	1,94	bar
cw.temp.engine outlet	86,7	°C	SW		
			sea water press.pump outlet	4,29	bar
			sea water temp.pump inlet	24,9	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	106,4	°C	°C	90,6	103,0	108,8	110,3	103,5	106,6	106,7	106,2	84,7
splash oil temperature	100,7	°C	°C		98,0	100,5	103,3	100,7	100,0	102,0		



MGLW

press.crankcase -8 mbar smoke AVL - -

optional values shown as "-" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 450,6 °C



21-09-2023, 11:15

85,0 %

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	24,7	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,79	g/kg
turbocharger no.	1343-02/-03	relative humidity	60	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	2.039,5	kW
engine speed	1.894	rpm
mean eff.press.	20,8	bar

Fuel

fuel temp.HP pump inlet	37	°C
fuel press.HP pump inlet	8,1	bar
fuel mass flow 1	415,7	kg/h
s.f.o.c (42700kJ/kg,ISO)	204,4	g/kWh

Common Rail

fuel index	88,2	%			
current feed duration A MV	1,7	ms	current feed duration B MV	1,7	ms
current feed timing A MV	11,2	°KW	current feed timing B MV	11,2	°KW

Charge Air

charge air temp. A	38,9	°C	air temp.compr.inlet B	33,2	°C
			charge air temp. B	38,3	°C
			charge air press. B	2.580	mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6			
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-			
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-			
exh.gas temp.turb inlet A	624,0	°C								exh.gas temp.turb inlet B	628,0	°C
exh.gas press.turb.outlet A	115	mbar								exh.gas press.turb.outlet B	115	mbar
TC.speed A	60.630	rpm								TC.speed B	60.560	rpm
wastegate position A	0	%								wastegate position B	0	%

Lubricating Oil

lub.oil press.engine inlet	4,5	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT			LT		
cw.press.engine inlet	1,81	bar	cw.press.IC LT inlet	2,06	bar
cw.temp.engine outlet	86,9	°C	SW		
			sea water press.pump outlet	4,59	bar
			sea water temp.pump inlet	24,3	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	108,5	°C	°C	91,5	104,7	111,1	112,9	105,1	108,6	108,8	108,4	85,1
splash oil temperature	102,4	°C	°C		99,5	101,9	105,2	102,2	101,5	104,1		



MAN

press.crankcase -9 mbar smoke AVL - -

optional values shown as "" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 468,6 °C

measurement-no: 65
FAT 100%

Performance Data



21-09-2023, 11:50

100,1 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,2	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,92	g/kg
turbocharger no.	1343-02/-03	relative humidity	58	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	2.403,0	kW
engine speed	1.999	rpm
mean eff.press.	23,2	bar

Fuel

fuel temp.HP pump inlet	38	°C
fuel press.HP pump inlet	8,2	bar
fuel mass flow 1	498,4	kg/h
s.f.o.c (42700kJ/kg,ISO)	207,9	g/kWh

Common Rail

fuel index	100,9	%
current feed duration A MV	1,9	ms
current feed timing A MV	13,5	°KW

current feed duration B MV	1,9	ms
current feed timing B MV	13,5	°KW

Charge Air

charge air temp. A	41,0	°C
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air temp.compr.inlet B	33,2	°C
charge air temp. B	40,5	°C
charge air press. B	2.850	mbar

Exhaust Gas

	MV	
temp. cylinder outlet (A)	-	°C
temp. cylinder outlet (B)	-	°C

cyl.	1	2	3	4	5	6
A	-	-	-	-	-	-
B	-	-	-	-	-	-

exh.gas temp.turb inlet A	692,0	°C
exh.gas press.turb.outlet A	137	mbar
TC.speed A	63.960	rpm
wastegate position A	9	%

exh.gas temp.turb inlet B	695,0	°C
exh.gas press.turb.outlet B	137	mbar
TC.speed B	63.830	rpm
wastegate position B	9	%

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT		
cw.press.engine inlet	1,91	bar
cw.temp.engine outlet	87,3	°C

LT		
cw.press.IC LT inlet	2,19	bar
SW		
sea water press.pump outlet	5,03	bar
sea water temp.pump inlet	24,9	°C

Bearing

	MV	
main bearing temperature	111,3	°C
splash oil temperature	105,3	°C

No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
°C	92,3	106,7	114,7	116,3	107,5	111,1	111,9	111,2	85,6
°C		103,2	104,5	108,4	104,9	104,0	107,1		



MAN

press.crankcase -8 mbar smoke AVL - -

optional values shown as "" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 523,6 °C



21-09-2023, 12:23

100,2 %

engine type	12V175D-ML	atmospheric pressure	995 mbar
engine no.	8351446	ambient temperature	25,8 °C
turbocharger type	TCR12-43063	abs. ambient humidity	11,84 g/kg
turbocharger no.	1343-02/-03	relative humidity	56 %
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40
testbed no.	5	fuel oil spec.	MGO
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg

Power

engine power	2.403,8 kW
engine speed	1.999 rpm
mean eff.press.	23,2 bar

Fuel

fuel temp.HP pump inlet	38 °C
fuel press.HP pump inlet	8,1 bar
fuel mass flow 1	498,1 kg/h
s.f.o.c (42700kJ/kg,ISO)	207,5 g/kWh

Common Rail

fuel index	100,8 %	current feed duration B MV	1,9 ms
current feed duration A MV	1,9 ms	current feed timing B MV	13,5 °KW
current feed timing A MV	13,5 °KW		

Charge Air

charge air temp. A	41,5 °C	air temp.compr.inlet B	35,4 °C
		charge air temp. B	40,8 °C
		charge air press. B	2.850 mbar

Exhaust Gas

	MV	cyl.	1	2	3	4	5	6
temp. cylinder outlet (A)	- °C	A	-	-	-	-	-	-
temp. cylinder outlet (B)	- °C	B	-	-	-	-	-	-
exh.gas temp.turb inlet A	692,0 °C							
exh.gas temp.turb inlet B	695,0 °C							
exh.gas press.turb.outlet A	135 mbar							
exh.gas press.turb.outlet B	135 mbar							
TC.speed A	64.100 rpm							
TC.speed B	63.860 rpm							
wastegate position A	8 %							
wastegate position B	8 %							

Lubricating Oil

lub.oil press.engine inlet	4,3 bar
lub.oil temp.engine inlet	85 °C

Cooling Water

HT

cw.press.engine inlet	1,91 bar
cw.temp.engine outlet	87,3 °C

LT

cw.press.IC LT inlet	2,19 bar
SW	
sea water press.pump outlet	4,88 bar
sea water temp.pump inlet	25,5 °C

Bearing

	MV	No.	supp. CS	1	2	3	4	5	6	7	CCS
main bearing temperature	111,4 °C	°C	92,5	106,8	114,7	116,4	107,5	111,2	111,9	111,1	85,6
splash oil temperature	105,5 °C	°C		103,5	104,7	108,6	105,0	104,1	107,2		



MAN

press.crankcase -9 mbar smoke AVL - -

optional values shown as "-" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 523,6 °C



21-09-2023, 12:40

110,0 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	26,2	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,99	g/kg
turbocharger no.	1343-02/-03	relative humidity	55	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990	kJ/kg

Power

engine power	2.639,8	kW
engine speed	2.064	rpm
mean eff.press.	24,7	bar

Fuel

fuel temp.HP pump inlet	38	°C
fuel press.HP pump inlet	8,1	bar
fuel mass flow 1	568,9	kg/h
s.f.o.c (42700kJ/kg,ISO)	215,8	g/kWh

Common Rail

fuel index	↑	112,0	%		
current feed duration A MV		2,1	ms	current feed duration B MV	2,1 ms
current feed timing A MV		14.2	°KW	current feed timing B MV	14,2 °KW

Charge Air

charge air temp. A	42,1	°C	air temp.compr.inlet B	35,2	°C
			charge air temp. B	41,5	°C
			charge air press. B	2.895	mbar

Exhaust Gas

	MV	cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	- °C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	- °C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	781,0 °C								exh.gas temp.turb inlet B 785,5 °C
exh.gas press.turb.outlet A	154 mbar								exh.gas press.turb.outlet B 154 mbar
TC.speed A	64.880 rpm								TC.speed B 64.740 rpm
wastegate position A	23 %								wastegate position B 23 %

Lubricating Oil

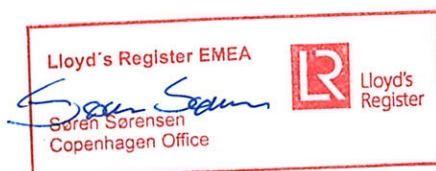
lub.oil press.engine inlet	4,2	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT			LT		
cw.press.engine inlet	1,98	bar	cw.press.IC LT inlet	2,29	bar
cw.temp.engine outlet	87,7	°C	SW		
			sea water press.pump outlet	5,27	bar
			sea water temp.pump inlet	25,6	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	112,7	°C	°C	92,8	107,6	116,3	118,2	108,7	112,3	113,0	112,5	85,9
splash oil temperature	107,2	°C	°C		105,1	106,5	110,5	106,5	105,5	109,0		



21-09-2023

SKEVVE

press.crankcase -9 mbar smoke AVL - -

optional values shown as "" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 613,2 °C

measurement-no: 68
FAT point 15 600rpm_2,7%

Performance Data



21-09-2023, 12:50

2,7 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	26,6	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,61	g/kg
turbocharger no.	1343-02/-03	relative humidity	52	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	64,7	kW
engine speed	599	rpm
mean eff.press.	2,1	bar

Fuel

fuel temp.HP pump inlet	40	°C
fuel press.HP pump inlet	6,9	bar
fuel mass flow 1	16,9	kg/h
s.f.o.c (42700kJ/kg,ISO)	-	g/kWh

Common Rail

fuel index	16,4	%		
current feed duration A MV	0,9	ms	current feed duration B MV	0,9 ms
current feed timing A MV	3,4	°KW	current feed timing B MV	3,4 °KW

Charge Air

charge air temp. A	39,5	°C	air temp.compr.inlet B	41,3 °C
			charge air temp. B	38,3 °C
			charge air press. B	10 mbar

Exhaust Gas

		MV	cyl.	1	2	3	4	5	6		
temp. cylinder outlet (A)		- °C	A	-	-	-	-	-	-		
temp. cylinder outlet (B)		- °C	B	-	-	-	-	-	-		
exh.gas temp.turb inlet A		298,8 °C					exh.gas temp.turb inlet B				296,3 °C
exh.gas press.turb.outlet A		-2 mbar					exh.gas press.turb.outlet B				-2 mbar
TC.speed A		7.630 rpm					TC.speed B				7.682 rpm
wastegate position A		0 %					wastegate position B				0 %

Lubricating Oil

lub.oil press.engine inlet	2,1	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	0,89	bar
cw.temp.engine outlet	86,4	°C

LT

cw.press.IC LT inlet	0,79	bar
----------------------	------	-----

SW

sea water press.pump outlet	1,33	bar
sea water temp.pump inlet	21,1	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	92,5	°C	°C	87,3	90,7	93,0	93,7	91,7	93,0	92,8	92,4	82,3
splash oil temperature	89,3	°C	°C		89,7	89,3	89,8	88,7	88,6	89,6		

press.crankcase

0 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 404,7 °C

measurement-no: 69
FAT point 16 900rpm_9,1%

Performance Data



21-09-2023, 12:57

9,1%

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	26,4	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,52	g/kg
turbocharger no.	1343-02/-03	relative humidity	53	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990	kJ/kg

Power

engine power	218,5	kW
engine speed	900	rpm
mean eff.press.	4,7	bar

Fuel

fuel temp.HP pump inlet	40	°C
fuel press.HP pump inlet	7,3	bar
fuel mass flow 1	48,0	kg/h
s.f.o.c (42700kJ/kg,ISO)	219,4	g/kWh

Common Rail

fuel index	27,2	%		
current feed duration A MV	1,0	ms	current feed duration B MV	1,0 ms
current feed timing A MV	3,7	°KW	current feed timing B MV	3,7 °KW

Charge Air

charge air temp. A	39,9	°C	air temp.compr.inlet B	43,4 °C
			charge air temp. B	39,0 °C
			charge air press. B	60 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	373,6	°C								exh.gas temp.turb inlet B 369,4 °C
exh.gas press.turb.outlet A	-1	mbar								exh.gas press.turb.outlet B -1 mbar
TC.speed A	13.320	rpm								TC.speed B 13.350 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	3,5	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT

cw.press.engine inlet	1,02	bar
cw.temp.engine outlet	86,7	°C

LT

cw.press.IC LT inlet 0,91 bar

SW

sea water press.pump outlet 1,78 bar
sea water temp.pump inlet 22,9 °C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	91,1	°C	°C	87,1	90,2	90,9	92,0	91,0	91,4	90,7	91,7	82,5
splash oil temperature	90,5	°C	°C		91,2	90,3	91,0	89,8	89,5	90,9		

press.crankcase

-0 mbar

smoke AVL

- -

optional values shown as ""

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 381,0 °C

measurement-no: 70
FAT point 17 600rpm_3,38%

Performance Data



21-09-2023, 13:03

3,4 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	26,1	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,45	g/kg
turbocharger no.	1343-02/-03	relative humidity	53	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	81,0	kW
engine speed	600	rpm
mean eff.press.	2,6	bar

Fuel

fuel temp.HP pump inlet	41	°C
fuel press.HP pump inlet	6,9	bar
fuel mass flow 1	19,8	kg/h
s.f.o.c (42700kJ/kg,ISO)	-	g/kWh

Common Rail

fuel index	18,8	%
current feed duration A MV	1,0	ms
current feed timing A MV	3,4	°KW

current feed duration B MV	1,0	ms
current feed timing B MV	3,4	°KW

Charge Air

charge air temp. A	40,3	°C	air temp.compr.inlet B	41,7	°C
			charge air temp. B	39,1	°C
			charge air press. B	0	mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-
exh.gas temp.turb inlet A	268,8	°C							
exh.gas press.turb.outlet A	-2	mbar							
TC.speed A	7.692	rpm							
wastegate position A	0	%							

exh.gas temp.turb inlet B	265,7	°C
exh.gas press.turb.outlet B	-2	mbar
TC.speed B	7.725	rpm
wastegate position B	0	%

Lubricating Oil

lub.oil press.engine inlet	2,1	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	0,89	bar
cw.temp.engine outlet	86,2	°C

LT

cw.press.IC LT inlet	0,80	bar
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SW

sea water press.pump outlet	1,33	bar
sea water temp.pump inlet	22,0	°C

Bearing

	MV	
main bearing temperature	89,9	°C
splash oil temperature	89,3	°C

No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
°C	86,3	89,0	89,9	90,7	90,0	90,2	89,8	90,0	82,0
°C		89,8	89,4	90,0	88,5	88,4	89,8		

press.crankcase 0 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 330,6 °C



8,0 %

21-09-2023, 13:08

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	26,0	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,49	g/kg
turbocharger no.	1343-02/-03	relative humidity	54	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	191,7	kW
engine speed	800	rpm
mean eff.press.	4,6	bar

Fuel

fuel temp.HP pump inlet	41	°C
fuel press.HP pump inlet	7,2	bar
fuel mass flow 1	41,8	kg/h
s.f.o.c (42700kJ/kg,ISO)	-	g/kWh

Common Rail

fuel index	27,1	%		
current feed duration A MV	1,0	ms	current feed duration B MV	1,0 ms
current feed timing A MV	3,5	°KW	current feed timing B MV	3,5 °KW

Charge Air

charge air temp. A	40,1	°C	air temp.compr.inlet B	41,5 °C
			charge air temp. B	39,1 °C
			charge air press. B	40 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	349,0	°C								exh.gas temp.turb inlet B 345,5 °C
exh.gas press.turb.outlet A	-1	mbar								exh.gas press.turb.outlet B -1 mbar
TC.speed A	11.540	rpm								TC.speed B 11.590 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	3,1	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT

cw.press.engine inlet	0,97	bar
cw.temp.engine outlet	86,8	°C

LT

cw.press.IC LT inlet	0,87	bar
SW		
sea water press.pump outlet	1,62	bar
sea water temp.pump inlet	22,1	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	90,2	°C	°C	86,4	89,3	90,1	90,9	90,3	90,4	90,0	90,4	82,0
splash oil temperature	90,4	°C	°C		91,1	90,4	91,0	89,6	89,4	90,8		

press.crankcase

-0 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 345,7 °C



21-09-2023, 13:14

15,6 %

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,8	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,39	g/kg
turbocharger no.	1343-02/-03	relative humidity	84	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	374,5	kW
engine speed	999	rpm
mean eff.press.	7,3	bar

Fuel

fuel temp.HP pump inlet	41	°C
fuel press.HP pump inlet	7,5	bar
fuel mass flow 1	79,8	kg/h
s.f.o.c (42700kJ/kg,ISO)	212,7	g/kWh

Common Rail

fuel index	37,4	%		
current feed duration A MV	1,1	ms	current feed duration B MV	1,1 ms
current feed timing A MV	3,8	°KW	current feed timing B MV	3,8 °KW

Charge Air

charge air temp. A	40,1	°C	air temp.compr.inlet B	43,3 °C
			charge air temp. B	39,3 °C
			charge air press. B	140 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	495,8	°C								exh.gas temp.turb inlet B 490,5 °C
exh.gas press.turb.outlet A	1	mbar								exh.gas press.turb.outlet B 1 mbar
TC.speed A	18.650	rpm								TC.speed B 18.650 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,1	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	1,07	bar
cw.temp.engine outlet	86,7	°C

LT

cw.press.IC LT inlet	0,97	bar
SW		
sea water press.pump outlet	2,01	bar
sea water temp.pump inlet	25,2	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	91,5	°C	°C	86,8	90,7	91,3	92,5	91,2	91,7	90,9	92,2	82,4
splash oil temperature	91,5	°C	°C		92,3	91,4	92,2	90,6	90,5	92,2		

press.crankcase

-0 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 436,1 °C

measurement-no: 73
FAT point 20_1170rpm_25%

Performance Data



21-09-2023, 13:19

25,0

% **PRELIMINARY**

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,7	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,40	g/kg
turbocharger no.	1343-02/-03	relative humidity	54	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	600,3	kW
engine speed	1.170	rpm
mean eff.press.	9,9	bar

Fuel

fuel temp.HP pump inlet	41	°C
fuel press.HP pump inlet	7,6	bar
fuel mass flow 1	124,8	kg/h
s.f.o.c (42700kJ/kg,ISO)	207,6	g/kWh

Common Rail

fuel index	49,1	%		
current feed duration A MV	1,1	ms	current feed duration B MV	1,1 ms
current feed timing A MV	4,4	°KW	current feed timing B MV	4,4 °KW

Charge Air

charge air temp. A	40,0	°C	air temp.compr.inlet B	43,0 °C
			charge air temp. B	39,3 °C
			charge air press. B	390 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	575,4	°C								exh.gas temp.turb inlet B 572,3 °C
exh.gas press.turb.outlet A	5	mbar								exh.gas press.turb.outlet B 5 mbar
TC.speed A	27.750	rpm								TC.speed B 27.690 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	1,18	bar
cw.temp.engine outlet	86,8	°C

LT

cw.press.IC LT inlet	1,06	bar
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SW

sea water press.pump outlet	2,37	bar
sea water temp.pump inlet	25,1	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	92,9	°C	°C	87,3	92,0	93,0	94,2	92,3	93,0	92,3	93,7	82,6
splash oil temperature	93,4	°C	°C		94,0	93,1	94,5	92,4	92,3	94,1		

press.crankcase

-0 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 492,5 °C

measurement-no: 74
FAT point 21_1270rpm_32%

Performance Data



21-09-2023, 13:25

31,9 %

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,5	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,42	g/kg
turbocharger no.	1343-02/-03	relative humidity	55	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	766,7	kW
engine speed	1.270	rpm
mean eff.press.	11,7	bar

Fuel

fuel temp.HP pump inlet	41	°C
fuel press.HP pump inlet	7,7	bar
fuel mass flow 1	155,9	kg/h
s.f.o.c (42700kJ/kg,ISO)	203,1	g/kWh

Common Rail

fuel index	56,1	%		
current feed duration A MV	1,2	ms	current feed duration B MV	1,2 ms
current feed timing A MV	5,0	°KW	current feed timing B MV	5,0 °KW

Charge Air

charge air temp. A	39,8	°C	air temp.compr.inlet B	43,0 °C
			charge air temp. B	39,1 °C
			charge air press. B	640 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	589,5	°C								591,3 °C
exh.gas press.turb.outlet A	11	mbar								11 mbar
TC.speed A	34.050	rpm								33.940 rpm
wastegate position A	0	%								0 %
exh.gas temp.turb inlet B										
exh.gas press.turb.outlet B										
TC.speed B										
wastegate position B										

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	1,25	bar
cw.temp.engine outlet	86,9	°C

LT

cw.press.IC LT inlet	1,11	bar
sea water press.pump outlet	2,62	bar
sea water temp.pump inlet	25,2	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	94,8	°C	°C	87,8	93,6	95,0	96,3	93,9	94,8	94,2	95,5	83,0
splash oil temperature	94,8	°C	°C		95,2	94,5	96,0	93,8	93,7	95,4		

press.crankcase

-1 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 510,8 °C

measurement-no: 75
FAT point 22_1372rpm_40,3%

Performance Data



21-09-2023, 13:30

40,3 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,6	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,50	g/kg
turbocharger no.	1343-02/-03	relative humidity	55	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	967,9	kW
engine speed	1.371	rpm
mean eff.press.	13,6	bar

Fuel

fuel temp.HP pump inlet	41	°C
fuel press.HP pump inlet	7,7	bar
fuel mass flow 1	191,7	kg/h
s.f.o.c (42700kJ/kg,ISO)	198,1	g/kWh

Common Rail

fuel index	60,2	%		
current feed duration A MV	1,3	ms	current feed duration B MV	1,3 ms
current feed timing A MV	7,7	°KW	current feed timing B MV	7,7 °KW

Charge Air

charge air temp. A	39,9	°C	air temp.compr.inlet B	39,7 °C
			charge air temp. B	39,1 °C
			charge air press. B	930 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	585,0	°C								587,0 °C
exh.gas press.turb.outlet A	19	mbar								19 mbar
TC.speed A	39.760	rpm								39.670 rpm
wastegate position A	0	%								0 %
exh.gas temp.turb inlet B										
exh.gas press.turb.outlet B										
TC.speed B										
wastegate position B										

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT

cw.press.engine inlet	1,33	bar
cw.temp.engine outlet	86,9	°C

LT

cw.press.IC LT inlet	1,18	bar
SW		
sea water press.pump outlet	2,90	bar
sea water temp.pump inlet	24,9	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	96,4	°C	°C	88,3	95,0	97,0	98,3	95,2	96,5	96,0	97,0	83,2
splash oil temperature	96,3	°C	°C		96,7	95,9	97,8	95,3	95,2	96,9		

press.crankcase

-2 mbar

smoke AVL

- -

optional values shown as ""

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 500,5 °C



21-09-2023, 13:35

50,0 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,6	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,42	g/kg
turbocharger no.	1343-02/-03	relative humidity	55	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	1.200,7	kW
engine speed	1.474	rpm
mean eff.press.	15,8	bar

Fuel

fuel temp.HP pump inlet	40	°C
fuel press.HP pump inlet	7,8	bar
fuel mass flow 1	233,6	kg/h
s.f.o.c (42700kJ/kg,ISO)	194,8	g/kWh

Common Rail

fuel index	64,0	%		
current feed duration A MV	1,4	ms	current feed duration B MV	1,4 ms
current feed timing A MV	10,1	°KW	current feed timing B MV	10,1 °KW

Charge Air

charge air temp. A	39,4	°C	air temp.compr.inlet B	37,3 °C
			charge air temp. B	38,6 °C
			charge air press. B	1.280 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	578,0	°C								578,0 °C
exh.gas press.turb.outlet A	30	mbar								30 mbar
TC.speed A	45.320	rpm								45.260 rpm
wastegate position A	0	%								0 %
exh.gas temp.turb inlet B										583,0 °C
exh.gas press.turb.outlet B										30 mbar
TC.speed B										45.260 rpm
wastegate position B										0 %

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT			LT		
cw.press.engine inlet	1,41	bar	cw.press.IC LT inlet	1,27	bar
cw.temp.engine outlet	86,8	°C	SW		
			sea water press.pump outlet	3,17	bar
			sea water temp.pump inlet	25,2	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	98,7	°C	°C	89,1	96,9	99,8	100,9	97,2	98,8	98,5	98,7	83,5
splash oil temperature	97,9	°C	°C		98,2	97,4	99,6	97,1	96,6	98,4		

press.crankcase -3 mbar smoke AVL - -

optional values shown as "-" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 481,0 °C

Performance Data



21-09-2023, 13:40

61,0 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,5	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,46	g/kg
turbocharger no.	1343-02/-03	relative humidity	55	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990	kJ/kg

Power

engine power	1.462,9	kW
engine speed	1.574	rpm
mean eff.press.	18,0	bar

Fuel

fuel temp.HP pump inlet	39	°C
fuel press.HP pump inlet	7,9	bar
fuel mass flow 1	282,8	kg/h
s.f.o.c (42700kJ/kg,ISO)	193,8	g/kWh

Common Rail

fuel index	71.8	%			
current feed duration A MV	1.5	ms	current feed duration B MV	1.5	ms
current feed timing A MV	11.6	°KW	current feed timing B MV	11.6	°KW

Charge Air

charge air temp. A	38,2	°C	air temp.compr.inlet B	35,1	°C
			charge air temp. B	37,4	°C
			charge air press. B	1.680	mbar

Exhaust Gas

	MV	cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	- °C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	- °C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	579,0 °C								exh.gas temp.turb inlet B 583,0 °C
exh.gas press.turb.outlet A	47 mbar								exh.gas press.turb.outlet B 47 mbar
TC.speed A	50.550 rpm								TC.speed B 50.530 rpm
wastegate position A	0 %								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,3	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT			LT		
			cw.press.IC LT inlet	1,48	bar
cw.press.engine inlet	1,50	bar	SW		
cw.temp.engine outlet	87,0	°C	sea water press.pump outlet	3,45	bar
			sea water temp.pump inlet	25,4	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	100,9	°C	°C	89,7	98,8	102,3	103,6	99,1	100,9	100,7	100,6	83,9
splash oil temperature	99,3	°C	°C		98,5	98,9	101,5	98,7	98,1	100,0		

press.crankcase -4 mbar smoke AVL - -

optional values shown as "-" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 464,2 °C



21-09-2023, 13:48

75,2 %

engine type	12V175D-ML	atmospheric pressure	995 mbar
engine no.	8351446	ambient temperature	25,5 °C
turbocharger type	TCR12-43063	abs. ambient humidity	11,50 g/kg
turbocharger no.	1343-02/-03	relative humidity	55 %
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40
testbed no.	5	fuel oil spec.	MGO
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg

Power

engine power	1.803,5	kW
engine speed	1.687	rpm
mean eff.press.	20,7	bar

Fuel

fuel temp.HP pump inlet	38 °C
fuel press.HP pump inlet	7,9 bar
fuel mass flow 1	351,7 kg/h
s.f.o.c (42700kJ/kg,ISO)	195,6 g/kWh

Common Rail

fuel index	84,0 %		
current feed duration A MV	1,7 ms	current feed duration B MV	1,7 ms
current feed timing A MV	11,0 °KW	current feed timing B MV	11,0 °KW

Charge Air

charge air temp. A	37,0 °C	air temp.compr.inlet B	34,2 °C
		charge air temp. B	36,5 °C
		charge air press. B	2.194 mbar

Exhaust Gas

	MV	cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	- °C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	- °C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	600,0 °C								603,0 °C
exh.gas press.turb.outlet A	72 mbar								72 mbar
TC.speed A	56.180 rpm								56.250 rpm
wastegate position A	0 %								0 %
exh.gas temp.turb inlet B									
exh.gas press.turb.outlet B									
TC.speed B									
wastegate position B									

Lubricating Oil

lub.oil press.engine inlet	4,4 bar
lub.oil temp.engine inlet	85 °C

Cooling Water

HT

cw.press.engine inlet	1,60 bar
cw.temp.engine outlet	87,1 °C

LT

cw.press.IC LT inlet	1,80 bar
SW	
sea water press.pump outlet	3,85 bar
sea water temp.pump inlet	25,1 °C

Bearing

	MV	No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	104,4 °C	°C	90,5	101,7	106,7	107,9	102,0	104,7	104,4	103,8	84,5
splash oil temperature	100,6 °C	°C		98,0	100,4	103,3	100,5	99,8	101,8		

press.crankcase -6 mbar smoke AVL - -

optional values shown as "" pressure values above atmosphere A / B = cylinder bank LT / HT = low / high temp. MV = mean value IC = intercooler

Remarks: exh. gas temp.turb outlet common 460,3 °C

Performance Data



21-09-2023, 13:54

85,0 %

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,4	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,53	g/kg
turbocharger no.	1343-02/-03	relative humidity	56	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	2.040,7	kW
engine speed	1.758	rpm
mean eff.press.	22,4	bar

Fuel

fuel temp.HP pump inlet	38	°C
fuel press.HP pump inlet	8,0	bar
fuel mass flow 1	404,3	kg/h
s.f.o.c (42700kJ/kg,ISO)	198,6	g/kWh

Common Rail

fuel index	92,8	%			
current feed duration A MV	1,9	ms	current feed duration B MV	1,9	ms
current feed timing A MV	11,1	°KW	current feed timing B MV	11,1	°KW

Charge Air

charge air temp. A	39,2	°C	air temp.compr.inlet B	33,8	°C
			charge air temp. B	38,5	°C
			charge air press. B	2.520	mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6			
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-			
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-			
exh.gas temp.turb inlet A	628,7	°C								exh.gas temp.turb inlet B	631,0	°C
exh.gas press.turb.outlet A	92	mbar								exh.gas press.turb.outlet B	92	mbar
TC.speed A	59.560	rpm								TC.speed B	59.560	rpm
wastegate position A	0	%								wastegate position B	0	%

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT

cw.press.engine inlet	1,66	bar
cw.temp.engine outlet	87,3	°C

LT

cw.press.IC LT inlet	1,88	bar
SW		
sea water press.pump outlet	4,07	bar
sea water temp.pump inlet	25,6	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	106,2	°C	°C	90,8	102,7	108,9	110,0	103,3	106,4	106,5	105,3	84,6
splash oil temperature	101,9	°C	°C		98,8	101,6	104,7	101,8	101,1	103,5		

press.crankcase

-8 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 473,3 °C

Performance Data



21-09-2023, 14:01

91,0 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,3	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,70	g/kg
turbocharger no.	1343-02/-03	relative humidity	57	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990	kJ/kg

Power

engine power	2.185,2	kW
engine speed	1.800	rpm
mean eff.press.	23,5	bar

Fuel

fuel temp.HP pump inlet	39	°C
fuel press.HP pump inlet	8,0	bar
fuel mass flow 1	435,5	kg/h
s.f.o.c (42700kJ/kg,ISO)	199,9	g/kWh

Common Rail

fuel index	97,8	%		
current feed duration A MV	1,9	ms	current feed duration B MV	1,9 ms
current feed timing A MV	10,7	°KW	current feed timing B MV	10,7 °KW

Charge Air

charge air temp. A	40,2	°C	air temp.compr.inlet B	33,2 °C
			charge air temp. B	39,4 °C
			charge air press. B	2.676 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	648,0	°C								exh.gas temp.turb inlet B 650,0 °C
exh.gas press.turb.outlet A	103	mbar								exh.gas press.turb.outlet B 103 mbar
TC.speed A	61.140	rpm								TC.speed B 61.170 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,3	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT

cw.press.engine inlet	1,71	bar
cw.temp.engine outlet	87,5	°C

LT

cw.press.IC LT inlet	1,92	bar
SW		
sea water press.pump outlet	4,24	bar
sea water temp.pump inlet	25,6	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	107,5	°C	°C	91,2	103,5	110,5	111,9	104,3	107,8	108,1	106,7	84,8
splash oil temperature	102,8	°C	°C		99,6	102,5	105,6	102,6	101,9	104,6		

press.crankcase

-9 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 486,0 °C

measurement-no: 81
FAT point 28_1700rpm_89,2%

Performance Data



21-09-2023, 14:06

89,2 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,5	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,84	g/kg
turbocharger no.	1343-02/-03	relative humidity	57	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	2.140,4	kW
engine speed	1.699	rpm
mean eff.press.	24,4	bar

Fuel

fuel temp.HP pump inlet	39	°C
fuel press.HP pump inlet	7,9	bar
fuel mass flow 1	418,6	kg/h
s.f.o.c (42700kJ/kg,ISO)	196,1	g/kWh

Common Rail

fuel index	100,1	%		
current feed duration A MV	1,9	ms	current feed duration B MV	1,9 ms
current feed timing A MV	10,3	°KW	current feed timing B MV	10,3 °KW

Charge Air

charge air temp. A	39,1	°C	air temp.compr.inlet B	34,1 °C
			charge air temp. B	38,2 °C
			charge air press. B	2.580 mbar

Exhaust Gas

	MV	cyl.	1	2	3	4	5	6	
			A	-	-	-	-	-	
temp. cylinder outlet (A)	- °C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	- °C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	646,5	°C							648,0 °C
exh.gas press.turb.outlet A	92	mbar							92 mbar
TC.speed A	59.840	rpm							59.900 rpm
wastegate position A	0	%							0 %
exh.gas temp.turb inlet B									
exh.gas press.turb.outlet B									
TC.speed B									
wastegate position B									

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	85	°C

Cooling Water

HT

cw.press.engine inlet	1,61	bar
cw.temp.engine outlet	87,4	°C

LT

cw.press.IC LT inlet	1,80	bar
SW		
sea water press.pump outlet	3,84	bar
sea water temp.pump inlet	24,8	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	CCS
main bearing temperature	106,8	°C	°C	91,1	103,2	109,7	111,0	103,7	107,2	107,2	105,8	84,7
splash oil temperature	102,3	°C	°C		99,1	102,0	105,2	102,3	101,4	103,9		

press.crankcase

-7 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 489,3 °C



21-09-2023, 14:11

76,5 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,3	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,74	g/kg
turbocharger no.	1343-02/-03	relative humidity	57	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	1.835,7	kW
engine speed	1.599	rpm
mean eff.press.	22,2	bar

Fuel

fuel temp.HP pump inlet	39	°C
fuel press.HP pump inlet	7,9	bar
fuel mass flow 1	351,9	kg/h
s.f.o.c (42700kJ/kg,ISO)	192,3	g/kWh

Common Rail

fuel index	89,1	%		
current feed duration A MV	1,8	ms	current feed duration B MV	1,8 ms
current feed timing A MV	11,2	°KW	current feed timing B MV	11,2 °KW

Charge Air

charge air temp. A	36,0	°C	air temp.compr.inlet B	34,9 °C
			charge air temp. B	35,0 °C
			charge air press. B	2.126 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	619,0	°C								exh.gas temp.turb inlet B 621,5 °C
exh.gas press.turb.outlet A	66	mbar								exh.gas press.turb.outlet B 66 mbar
TC.speed A	55.360	rpm								TC.speed B 55.440 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	1,52	bar
cw.temp.engine outlet	86,7	°C

LT

cw.press.IC LT inlet	1,69	bar
SW		
sea water press.pump outlet	3,56	bar
sea water temp.pump inlet	23,9	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	104,6	°C	°C	90,7	101,7	106,9	108,3	102,0	105,1	104,8	103,7	84,2
splash oil temperature	100,8	°C	°C		99,1	100,5	103,4	100,5	99,7	101,9		

press.crankcase

-5 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 482,8 °C



21-09-2023, 14:17

65,2 %

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,4	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,69	g/kg
turbocharger no.	1343-02/-03	relative humidity	57	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	1.565,7	kW
engine speed	1.499	rpm
mean eff.press.	20,2	bar

Fuel

fuel temp.HP pump inlet	38	°C
fuel press.HP pump inlet	7,8	bar
fuel mass flow 1	299,1	kg/h
s.f.o.c (42700kJ/kg,ISO)	191,5	g/kWh

Common Rail

fuel index	80,4	%		
current feed duration A MV	1,7	ms	current feed duration B MV	1,7 ms
current feed timing A MV	11,4	°KW	current feed timing B MV	11,4 °KW

Charge Air

charge air temp. A	35,7	°C	air temp.compr.inlet B	35,6 °C
			charge air temp. B	34,7 °C
			charge air press. B	1.740 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	609,0	°C								exh.gas temp.turb inlet B 612,0 °C
exh.gas press.turb.outlet A	47	mbar								exh.gas press.turb.outlet B 47 mbar
TC.speed A	51.110	rpm								TC.speed B 51.180 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,3	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	1,43	bar
cw.temp.engine outlet	86,8	°C

LT

cw.press.IC LT inlet	1,46	bar
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SW

sea water press.pump outlet	3,30	bar
sea water temp.pump inlet	24,0	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	101,9	°C	°C	90,1	99,5	103,8	105,0	99,7	102,3	102,1	101,1	83,9
splash oil temperature	99,7	°C	°C		99,8	99,1	101,7	99,0	98,3	100,4		

press.crankcase	-3	mbar	smoke AVL	-	-
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measurement-no: 84
FAT point 31_1400rpm_55,4%

Performance Data



21-09-2023, 14:23

56,1 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,4	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,68	g/kg
turbocharger no.	1343-02/-03	relative humidity	57	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	1.345,9	kW
engine speed	1.399	rpm
mean eff.press.	18,6	bar

Fuel

fuel temp.HP pump inlet	38	°C
fuel press.HP pump inlet	7,8	bar
fuel mass flow 1	259,2	kg/h
s.f.o.c (42700kJ/kg,ISO)	192,7	g/kWh

Common Rail

fuel index	74,3	%		
current feed duration A MV	1,6	ms	current feed duration B MV	1,6 ms
current feed timing A MV	9,9	°KW	current feed timing B MV	9,9 °KW

Charge Air

charge air temp. A	37,9	°C	air temp.compr.inlet B	38,4 °C
			charge air temp. B	36,5 °C
			charge air press. B	1.420 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	620,0	°C								exh.gas temp.turb inlet B 623,0 °C
exh.gas press.turb.outlet A	33	mbar								exh.gas press.turb.outlet B 33 mbar
TC.speed A	47.070	rpm								TC.speed B 47.120 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT			LT		
cw.press.engine inlet	1,35	bar	cw.press.IC LT inlet	1,26	bar
cw.temp.engine outlet	86,7	°C	SW		
			sea water press.pump outlet	2,98	bar
			sea water temp.pump inlet	24,2	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	99,7	°C	°C	89,4	97,6	101,2	102,1	97,8	100,0	99,7	99,3	83,6
splash oil temperature	98,4	°C	°C		98,8	97,9	100,0	97,5	97,0	99,1		

press.crankcase

-2 mbar

smoke AVL

- -

optional values shown as ""

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 509,4 °C

measurement-no: 85
FAT point 32_1350rpm_51%

Performance Data



21-09-2023, 14:30

50,9

%

PRELIMINARY

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,3	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,55	g/kg
turbocharger no.	1343-02/-03	relative humidity	56	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	1.222,7	kW
engine speed	1.350	rpm
mean eff.press.	17,5	bar

Fuel

fuel temp.HP pump inlet	38	°C
fuel press.HP pump inlet	7,8	bar
fuel mass flow 1	237,2	kg/h
s.f.o.c (42700kJ/kg,ISO)	194,1	g/kWh

Common Rail

fuel index	70,6	%		
current feed duration A MV	1,5	ms	current feed duration B MV	1,5 ms
current feed timing A MV	8,6	°KW	current feed timing B MV	8,6 °KW

Charge Air

charge air temp. A	38,3	°C	air temp.compr.inlet B	39,3 °C
			charge air temp. B	37,4 °C
			charge air press. B	1.240 mbar

Exhaust Gas

	MV		cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	-	°C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	-	°C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	628,0	°C								exh.gas temp.turb inlet B 631,0 °C
exh.gas press.turb.outlet A	26	mbar								exh.gas press.turb.outlet B 26 mbar
TC.speed A	44.590	rpm								TC.speed B 44.540 rpm
wastegate position A	0	%								wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT			LT		
cw.press.engine inlet	1,31	bar	cw.press.IC LT inlet	1,20	bar
cw.temp.engine outlet	86,7	°C	SW		
			sea water press.pump outlet	2,82	bar
			sea water temp.pump inlet	24,5	°C

Bearing

	MV		No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	98,2	°C	°C	88,8	96,2	99,4	100,3	96,5	98,5	98,1	98,0	83,3
splash oil temperature	97,5	°C	°C		97,9	97,1	99,0	96,6	96,2	98,3		

press.crankcase -1 mbar

smoke AVL

- -

optional values shown as ""

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 525,1 °C

measurement-no: 86
FAT point 33_1300rpm_46,8%

Performance Data



21-09-2023, 14:37

46,8 %

engine type	12V175D-ML	atmospheric pressure	995	mbar
engine no.	8351446	ambient temperature	25,4	°C
turbocharger type	TCR12-43063	abs. ambient humidity	11,60	g/kg
turbocharger no.	1343-02/-03	relative humidity	56	%
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40	
testbed no.	5	fuel oil spec.	MGO	
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg	

Power

engine power	1.124,1	kW
engine speed	1.300	rpm
mean eff.press.	16,7	bar

Fuel

fuel temp.HP pump inlet	38	°C
fuel press.HP pump inlet	7,7	bar
fuel mass flow 1	220,2	kg/h
s.f.o.c (42700kJ/kg,ISO)	196,0	g/kWh

Common Rail

fuel index	68,5	%		
current feed duration A MV	1,4	ms	current feed duration B MV	1,4 ms
current feed timing A MV	7,2	°KW	current feed timing B MV	7,2 °KW

Charge Air

charge air temp. A	38,7	°C	air temp.compr.inlet B	39,5	°C
			charge air temp. B	37,8	°C
			charge air press. B	1.100	mbar

Exhaust Gas

	MV	cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	- °C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	- °C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	640,0	°C							exh.gas temp.turb inlet B 644,0 °C
exh.gas press.turb.outlet A	22	mbar							exh.gas press.turb.outlet B 22 mbar
TC.speed A	42.370	rpm							TC.speed B 42.280 rpm
wastegate position A	0	%							wastegate position B 0 %

Lubricating Oil

lub.oil press.engine inlet	4,4	bar
lub.oil temp.engine inlet	84	°C

Cooling Water

HT

cw.press.engine inlet	1,27	bar
cw.temp.engine outlet	86,7	°C

LT

cw.press.IC LT inlet	1,15	bar
----------------------	------	-----

SW

sea water press.pump outlet	2,67	bar
sea water temp.pump inlet	24,4	°C

Bearing

	MV	No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	97,0 °C	°C	88,3	95,1	98,1	99,0	95,6	97,3	96,8	97,0	82,9
splash oil temperature	96,7 °C	°C		97,1	96,3	98,1	95,7	95,4	97,5		

press.crankcase

-1 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 542,9 °C



21-09-2023, 14:46

39,5 %

engine type	12V175D-ML	atmospheric pressure	995 mbar
engine no.	8351446	ambient temperature	25,5 °C
turbocharger type	TCR12-43063	abs. ambient humidity	11,50 g/kg
turbocharger no.	1343-02/-03	relative humidity	55 %
attached pumps	5 -	lube oil spec.	Shell Rimula R6 MS 10W-40
testbed no.	5	fuel oil spec.	MGO
water brake type	Horiba DT6300-2	LHV fuel	42.990 kJ/kg

Power

engine power	947,1 kW
engine speed	1.199 rpm
mean eff.press.	15,3 bar

Fuel

fuel temp.HP pump inlet	39 °C
fuel press.HP pump inlet	7,6 bar
fuel mass flow 1	190,8 kg/h
s.f.o.c (42700kJ/kg,ISO)	201,4 g/kWh

Common Rail

fuel index	64,6 %	current feed duration B MV	1,3 ms
current feed duration A MV	1,3 ms	current feed timing B MV	5,0 °KW
current feed timing A MV	5,0 °KW		

Charge Air

charge air temp. A	38,9 °C	air temp.compr.inlet B	41,5 °C
		charge air temp. B	38,2 °C
		charge air press. B	820 mbar

Exhaust Gas

	MV	cyl.	1	2	3	4	5	6	
temp. cylinder outlet (A)	- °C	A	-	-	-	-	-	-	
temp. cylinder outlet (B)	- °C	B	-	-	-	-	-	-	
exh.gas temp.turb inlet A	671,0 °C								676,6 °C
exh.gas press.turb.outlet A	14 mbar								14 mbar
TC.speed A	37.720 rpm								37.620 rpm
wastegate position A	0 %								0 %
exh.gas temp.turb inlet B									
exh.gas press.turb.outlet B									
TC.speed B									
wastegate position B									

Lubricating Oil

lub.oil press.engine inlet	4,4 bar
lub.oil temp.engine inlet	83 °C

Cooling Water

HT

cw.press.engine inlet	1,19 bar
cw.temp.engine outlet	86,8 °C

LT

cw.press.IC LT inlet	1,08 bar
SW	
sea water press.pump outlet	2,43 bar
sea water temp.pump inlet	24,6 °C

Bearing

	MV	No.	supp. CS	1	2	3	4	5	6	7	supp. CCS
main bearing temperature	95,1 °C	°C	87,5	93,4	96,1	96,9	93,7	95,5	94,9	95,3	82,5
splash oil temperature	95,4 °C	°C		95,9	95,0	96,6	94,3	94,1	96,2		

press.crankcase

-1 mbar

smoke AVL

- -

optional values shown as "-"

pressure values above atmosphere

A / B = cylinder bank

LT / HT = low / high temp.

MV = mean value

IC = intercooler

Remarks: exh. gas temp.turb outlet common 583,4 °C

Engine type 12V175DML

Engine no 8351446

Software and display versions

PRELIMINARY

		Versions / Date / Type			
		Marine Application			
Overload [h]	0	-CM-	-GM G-Safety-	-CM-	-GM G-Safety-
	0	-SatSys-	-SatSys-	-E.C. AISys-	-E.C. AISys-
Version		1.02.04.	1.02.04.	1.02.04.	1.02.04.
Setting		23-06-29	70-01-01	23-06-29	23-06-29
IM 1-					
on (-IM-)		1170			
Setting (-IM-)		23-06-29			
Version	3.01.13				
12 Cyl.	175D_V12CR				
8351446					

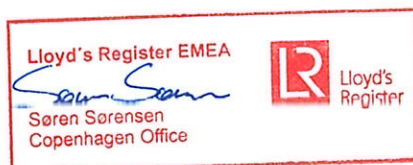
Legend:

- SatSys- Safety System
- CM = Control Module
- GM = Gateway Module
- IM = Injection Module
- E.C. AISys- Engine Control System
- Alarmproc. Unit

Parameters / Emission-Identifier

Parameters	Emission-Identifier
	1_2 00 . 00 . 00 . 00 . 0 0000 . 00 2_2 0000 . 0000
0,00 *Crank Delivery Begin Offset	F0C63913 5DAC0048

Engine n°: 8351446



CPN2330412/2

Remark: Actual running hours after FAT may differ from the value above.